

The Digitization of Tacit Knowledge: CIAER as a Candidate Minimum Sufficient Schema for Adaptive Purposeful Behavior and a Framework for Capturing Human Expertise in Cyber-Physical Systems

Kahn Capps

Capps Family Enterprises · Helena-West Helena, Arkansas

2026

Abstract

This paper presents a framework for the systematic digitization of human expertise within cyber-physical systems, grounded in a central theoretical proposition: the five-phase decision event schema introduced here — Cause, Intuition, Action, Effect, Result (CIAER) — is proposed as a candidate minimum sufficient behavioral schema for adaptive purposeful behavior in embodied agents. We propose that CIAER describes the protocol by which any embodied agent manages the continuous bidirectional reconciliation between an internal world model and an external environment: the outward arc flowing from registered environmental discrepancy through model inference and action to environmental restructuring, and the inward arc flowing from the restructured environment through outcome evaluation back to updated model state. The knowledge graph constructed from accumulated CIAER events is accordingly characterized as a partial externalization of the bidirectional reconciliation history between a specific agent and a specific environment — the accumulated record of every prediction error registered, every action selected, and every model update performed — largely inaccessible through conventional documentation approaches because the underlying knowledge was never encoded in propositional form.

We situate this proposition within a lineage of convergent frameworks drawn from seven disciplines, some mutually influential and some genuinely independent: Dewey's sensorimotor coordination circuit (1896), Wiener's cybernetics and Bateson's organism-environment mind theory (1948, 1972), Polanyi's epistemology of tacit knowledge (1958, 1966), Nonaka and Takeuchi's SECI model (1995), Dreyfus and Dreyfus's expert skill acquisition model (1986), Rasmussen's Skill-Rule-Knowledge hierarchy (1983), Klein's Recognition-Primed Decision model (1993), Endsley's Situation Awareness framework (1995), Boyd's OODA Loop (1986), Kolb's Experiential Learning Cycle (1984), and Friston's Free Energy Principle and Active Inference framework (2005–present), supplemented by Damasio's Somatic Marker Hypothesis (1994) as the neurobiological grounding for the biometric capture architecture. Each tradition describes, from a different disciplinary vantage, a behavioral structure functionally isomorphic to CIAER. We argue that this convergence — particularly at the fine-grained level of a specific five-phase decomposition, not merely the shared cyclic structure — constitutes strong grounds for the proposition, while acknowledging that the proposition itself remains a hypothesis requiring direct empirical validation. The minimality of the schema is advanced as a testable prediction: if an embodied agent can be shown to act adaptively while systematically

omitting one of the five phases, or if an alternative decomposition with fewer phases can describe the same behavioral space, the minimality claim is falsified.

We integrate the framework with the causal confusion literature in imitation learning, which we argue supplies its strongest technical validation: purely behavioral demonstrations systematically confuse correlates of expert action with causes, and the externalization bottleneck of tacit knowledge is precisely the bottleneck through which causal structure fails to transmit. CIAER's explicit decomposition of each event into Cause, Intuition, Action, Effect, and Result supplies the causal scaffolding that behavioral cloning lacks.

We apply this framework to the accelerating industrial crisis of cognitive capital depletion and propose a layered multimodal capture architecture comprising AI-glasses-integrated visual and acoustic sensors, wearable biometric sensing grounded in Damasio's somatic marker theory, and a three-stage attention capture pipeline. A proof-of-concept implementation, ArcShield, validates the foundational pipeline on a live PVC extrusion line in Helena-West Helena, Arkansas. As of April 2026, the system has captured initial decision events during active production, with biometric baseline calibration in progress and gaze dwell detection operational. Voice annotation capture has been validated in test sessions; the structured five-phase schema is functional and generating CIAER event records on a live production floor.

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1. The Macroeconomic Crisis of Cognitive Capital Depletion

The global industrial and operational landscape is currently undergoing a severe demographic contraction. The Alliance for Lifetime Income's Retirement Income Institute identifies 2024 through 2027 as the "Peak 65 Zone" — a period during which over 11,200 Americans reach traditional retirement age every single day, the highest sustained rate in the nation's demographic history, with 4.18 million Americans turning 65 in 2025 alone. Within United States manufacturing, approximately 26% of the workforce is aged 55 or older, and the average effective retirement age has declined to 61 years. Deloitte and The Manufacturing Institute project a requirement for 3.8 million new manufacturing employees by 2033, with 1.9 million positions at risk of remaining unfilled due to severe skills gaps.

Industry estimates attribute approximately \$92 billion in annual costs to human error-related downtime, inefficiency, and knowledge management failures directly tied to this demographic exodus. The qualitative dimension is equally severe: 70% of critical operational knowledge is tribal and never documented in any form. Research by Interloom found a roughly 50% gap between documented and actual operational knowledge even in enterprises that consider themselves well-documented. When veteran operators retire, the performance differential between best and average shifts widens dramatically, and routine troubleshooting devolves into costly trial-and-error.

The conventional framing of this crisis — as a documentation problem correctable through improved SOP generation or better knowledge management software — reflects a category error about the nature of the knowledge being lost. The tacit expertise of a veteran industrial operator extends substantially beyond the rule-like knowledge that documentation approaches assume it to consist of. It is, as the theoretical framework developed in Section 2 demonstrates, the accumulated bidirectional reconciliation history between an individual agent and a specific operational environment, encoded in the non-propositional parameter-state of a learned generative model. It cannot be reconstructed from explicit documentation because it was never encoded in propositional form in the first place, and the documentation process can only access the propositional outputs of a model whose essential nature is non-propositional.

2. Theoretical Foundations: Convergent Evidence for a Bidirectional Schema

The theoretical foundation of this paper rests on a convergence across eleven frameworks drawn from seven academic disciplines over a period spanning 130 years. We do not claim that these traditions are entirely independent — several have influenced one another across disciplinary boundaries, as we acknowledge below — but we do argue that the structural equivalence of their descriptions constitutes strong evidence for the existence of a real underlying phenomenon they are each attempting to describe. The CIAER schema names that phenomenon and proposes it as a candidate minimum sufficient behavioral schema for adaptive purposeful behavior in embodied agents: the smallest complete description of what an agent must do to act purposefully in a dynamic environment.

This proposition is explicitly advanced as a hypothesis, not a theorem. It is grounded in convergent theoretical testimony and consistent with neurobiological findings on prediction error processing and somatic marking, but it requires direct empirical validation in the industrial context through the research program described in Sections 5 and 8. We make no claim to have proved the hypothesis; we claim to have assembled sufficient grounds to take it seriously and to build a practical knowledge capture architecture upon it.

2.1 The Foundational Claim: Candidate Minimum Sufficient Schema for Adaptive Purposeful Behavior

An embodied agent acting purposefully in a dynamic environment must, at minimum: (1) detect that its current environmental state deviates from an expected state — *Cause*; (2) perform the internal computation that transforms that prediction error into an action selection — *Intuition*; (3) execute the selected action, crossing the agent-environment boundary outward — *Action*; (4) receive the environmental response — *Effect*; and (5) integrate that response back across the boundary to update the internal model — *Result*. We propose that an agent omitting any of these phases produces a specific, named failure mode: omitting Cause produces failure to detect environmental states requiring action; omitting Intuition produces stimulus-response rigidity without model-based inference; omitting Action produces paralysis; omitting Effect produces ungrounded inference

disconnected from environmental feedback; omitting Result produces failure to learn from outcomes.

CIAER is accordingly proposed as a candidate minimum complete description of purposeful behavior, supported by the sufficiency argument above and by the convergent evidence presented in Sections 2.2–2.14. The minimality of this decomposition is advanced as a testable prediction of the hypothesis, not as a demonstrated property. If an embodied agent can be shown to act adaptively while systematically omitting one of the five phases, or if an alternative decomposition with fewer phases can describe the same behavioral space without loss of capture capability, the minimality claim is falsified.

2.1.1 Topological Justification: Phases as Boundary-Crossings

The five-phase count is not an arbitrary subdivision of cognition but follows from the topology of bidirectional agent-environment reconciliation. The phases are individuated by the direction of information flow and the boundary-crossings between agent and environment, not by enumeration of internal micro-operations. *Cause* is the inward boundary-crossing in which environmental information registers as a prediction error against the agent's model. *Intuition* is the internal state transformation that occurs entirely on the agent side of the boundary, transforming the registered error into an action selection. *Action* is the outward boundary-crossing in which the agent transmits information to the environment by restructuring it. *Effect* is the environment's response state, which exists independently of whether the agent is currently observing it. *Result* is the second inward boundary-crossing in which the agent integrates the environmental response into updated model state. Five boundary-crossings and state-transitions yield five phases. Alternative decompositions with fewer phases collapse distinct boundary-crossings (as in classical stimulus-response theory, which fuses Intuition with Action) or eliminate the distinction between environmental response and model update (as in aggregate cybernetic feedback models, which fuse Effect with Result). Both collapses eliminate capture capabilities essential to the framework's purpose: collapsing Intuition and Action loses the ability to capture the inference that selected the action; collapsing Effect and Result loses the ability to distinguish confirming from disconfirming outcomes.

Critically, internal sub-structure within any phase does not generate additional phases. A chess grandmaster performing Klein's Variation 3 Action Evaluation generates candidate moves, simulates their consequences, and selects among them — three sub-computations that all occur on the same side of the agent-environment boundary, between receiving the position and committing to a move. These are rich internal structure within the Intuition phase, not separate phases, because none of them crosses a new boundary. Similarly, the Cause phase may internally contain detection, classification, and prioritization sub-processes, and the Result phase may internally contain outcome evaluation, model update, and memory encoding sub-processes. Phase count is determined by topology; phase richness is determined by the sophistication of the agent's internal machinery.

2.1.2 Multi-Timescale Tolerance: Phases Need Not Operate on the Same Clock

The CIAER phases are not required to operate on uniform timescales within a given agent. In human expert cognition during industrial operation, the full Cause-to-Result cycle typically completes within seconds to minutes. In the Venus flytrap, Cause (hair deflection) and Intuition (calcium ion integration across trigger hairs) occur within seconds; Action (trap closure) completes in approximately 100 milliseconds once triggered; Effect (prey capture or false alarm) is registered across minutes; Result (commit to digestion or reopen the trap) unfolds across hours; and a longer Result (organ-level sensitivity calibration based on accumulated closure history) unfolds across weeks. The phases span six orders of magnitude in timescale within a single agent. In the patellar reflex, Cause through Action complete in under 50 milliseconds through the monosynaptic spinal arc, while Result is distributed across multiple slower mechanisms: gamma motor neuron gain modulation adjusts spindle sensitivity across seconds to minutes, descending inputs from higher centers suppress or enhance the reflex across behavioral episodes, and long-term plasticity alters reflex thresholds across years of training or injury. The reflex's outward arc is fast and its inward arc is slow, but both arcs are present.

The implication is that the framework tolerates heterogeneous phase timescales within a single agent. A system with fast Cause-Intuition-Action-Effect and slow Result is still a complete CIAER loop — it simply implements the inward arc on a different clock than the outward arc, through separate anatomical or computational channels. This is not a loosening of the framework; it is a clarification of what the framework actually requires. The requirement is topological presence of all five phases, not temporal coupling between them.

2.1.3 Scope: What Counts as an Embodied Agent

The framework applies to embodied agents, and the defining characteristic of an embodied agent requires explicit specification. We propose a two-level criterion: a functional criterion that can be tested through observation, and a grounding criterion that explains why the functional criterion picks out anything meaningful.

Functional criterion. An embodied agent is a physical system that maintains its own organization against environmental perturbation through bidirectional reconciliation between an internal model and an external environment, possessing some mechanism of experience-dependent model update operating at any timescale. This criterion admits the Venus flytrap (calcium integration discriminates prey from noise; sensitivity calibrates across closure history), the bacterium performing chemotaxis (receptor methylation adapts to recent concentration gradients), the patellar reflex considered as a sub-system within the larger organism (multi-timescale plasticity mechanisms), the expert welder (somatic pattern recognition built from decades of experience), and the chess grandmaster (explicit model of positional dynamics updated by tournament outcomes). It excludes the bimetallic-strip thermostat, which has no experience-dependent update mechanism at any timescale, and the rolling rock, which has no self-maintenance orientation.

Grounding criterion. The functional criterion alone admits too many systems. Any thermostat with a re-calibration screw, on a sufficiently permissive reading, might count as

updating its model. What the framework requires is not mere parameter adjustment but adjustment *in the service of* the system's continued organization against perturbation — which requires the system to have something at stake in the reconciliation. We ground this in Millikan's teleosemantic criterion: the agent's representations have *content* because they were produced by a history of selection — evolutionary, developmental, or learning-based — that favored accurate models over inaccurate ones. The sterile welder whose reproductive history is complete, or the vasectomized operator with decades of acquired expertise, is unambiguously an embodied agent because his agency was *established* by the selection history that built him, even though his individual reproductive future is moot. An AI system, by contrast, has only derivative intentionality: its representations have content because they were shaped by an optimization process whose objectives were supplied by biological agents. This distinction is philosophically consequential and practically irrelevant: the commercial goal of transferring expertise from operator to AI is unaffected by the observation that the two agencies have different natural histories.

2.2 Dewey's Reflex Arc Critique: The Founding Bidirectional Insight

John Dewey's 1896 paper "The Reflex Arc Concept in Psychology" is the earliest articulation of the bidirectional reconciliation thesis in Western thought. Dewey argued that the prevailing stimulus-response framework misrepresented behavior by treating stimulus, central process, and response as serially independent events. His central insight was that the act of perceiving — of selecting what counts as stimulus — is itself an action of the organism, shaped by what the organism is currently doing and what it expects. Perception and action are not bookends of a linear arc but co-constitutive poles of a continuous coordinative circuit. The "stimulus" is not a given but is produced through the organism's ongoing engagement with the environment; the "response" reconfigures what the next stimulus will be.

This is structurally the bidirectional reconciliation CIAER formalizes. Dewey's coordinative circuit maps cleanly onto the outward and inward arcs: the outward arc corresponds to Dewey's "response" side — the organism acting on the environment — and the inward arc to his "sensorimotor coordination" — the organism integrating the consequences of that action back into its own state, which in turn shapes the next perception. Dewey did not decompose the circuit into five phases; he argued for the inadequacy of three (S-R-central-process). The CIAER phases add the boundary-crossing resolution that Dewey's critique demands but does not itself supply.

2.3 Cybernetics and Bateson's Organism-Environment Mind

Norbert Wiener's *Cybernetics* (1948) formalized feedback-regulated systems mathematically. A cybernetic system comprises a controller, a plant, a sensor, and a feedback path that returns information about the plant's state to the controller. Wiener's contribution was to demonstrate that this structure is invariant across biological, mechanical, and social systems — that the mathematics of negative feedback describe homeostasis in organisms, servomechanisms in engineering, and price adjustment in markets with equal force. Gregory Bateson's extension in *Steps to an Ecology of Mind* (1972) argued that cognition is not contained within the skull but distributed across the

agent-environment circuit itself; what we call “mind” is the closed feedback loop, and attempting to locate mind exclusively inside the organism commits a category error analogous to locating a clap inside one of the two hands.

The cybernetic tradition converges on the bidirectional reconciliation thesis but, like Dewey, does not resolve to five phases. Its characteristic decomposition is three (sensor, controller, actuator) or four (controller, actuator, environment, sensor). The CIAER refinement is to split the controller into Intuition (the model-based inference) and Result (the model update), and to distinguish Effect (the environmental state) from Result (the agent’s registration of that state). These refinements preserve the cybernetic insight while recovering capture capabilities that collapsed decompositions lose.

2.4 Polanyi’s Tacit Dimension: The Structural Inarticulability

Michael Polanyi’s *Personal Knowledge* (1958) and *The Tacit Dimension* (1966) advanced the thesis that “we can know more than we can tell.” Polanyi’s examples — face recognition, bicycle riding, expert medical diagnosis — share the feature that the practitioner performs reliably but cannot articulate the rules by which performance proceeds. Polanyi argued that this is not a failure of articulation but a *structural property* of skilled knowing: the skill consists precisely in the integration of subsidiary particulars into focal performance, and articulating the particulars decomposes the integration that constituted the skill. Asking the expert to articulate is asking him to dissociate the integrated performance back into its components, which cannot be done without destroying the skill under examination.

Polanyi’s tacit dimension is what CIAER’s externalization approach proposes to capture *through observation rather than articulation*. The full CIAER event record — environmental Cause with gaze dwell, biometric Intuition with EDA/HRV signature, observed Action with machine telemetry, measured Effect, and confirmed or disconfirmed Result — records the expert’s skilled performance *as it operates*, without demanding verbal decomposition. Voice annotation is supplementary, not foundational; the event is captured through its behavioral and physiological signatures even when the expert cannot or does not narrate it.

2.5 Argote and Ingram on Knowledge Transfer

Argote and Ingram (2000) demonstrated empirically that knowledge transfer between organizational units fails at predictable rates when the knowledge in question is embedded in the practices of specific people, tools, and their interaction. Their framework distinguishes knowledge-in-people, knowledge-in-tools, and knowledge-in-practice, and predicts that units sharing people transfer knowledge successfully, units sharing tools transfer moderately, and units sharing only documents transfer poorly. This prediction has been repeatedly confirmed across industries. The CIAER externalization architecture is designed to convert knowledge-in-people into knowledge-in-a-structured-corpus without requiring the transfer of the people themselves — a move that Argote and Ingram’s framework predicts will be difficult and which requires the specific capture technologies proposed in Section 5 to become tractable.

2.6 The SECI Model: The Externalization Bottleneck

Nonaka and Takeuchi's SECI model (1995) provides the definitive organizational framework for understanding how knowledge is generated, converted, and transferred within organizations (Argote and Ingram, 2000). The critical failure point in modern industry is the Externalization phase — the conversion of tacit expertise into explicit, scalable formats. Conventional approaches to Externalization attempt to translate the non-propositional generative model into propositional content, losing everything that makes the model intelligent in the translation: its context-sensitivity, its temporal dynamics, and its calibrated prediction functions. The CIAER architecture performs Externalization through behavioral and physiological observation of the model's operation, preserving the dynamic structure that propositional translation destroys.

2.7 Dreyfus and Dreyfus: The Expert's Transition to Tacit Performance

Hubert and Stuart Dreyfus's model of skill acquisition (*Mind Over Machine*, 1986) provides empirical grounding for the SRK hierarchy's treatment of expert cognition and the CIAER framework's approach to Skill-based event capture. The Dreyfus model identifies five developmental stages through which practitioners progress: novice, advanced beginner, competent, proficient, and expert. The critical transition occurs at the expert stage: experts act intuitively, without reflective decision-making. As the comprehensive review in Ericsson (2006) confirms, expertise involves vast repertoires of prior experiences enabling immediate perception of situational demands, without analytical computation.

The Dreyfus model directly grounds the CIAER framework's most important practical problem: the very expert behavior that is most valuable to capture is the behavior that experts are least able to report. The novice follows explicit rules and can articulate them; the expert responds intuitively to situational gestalts and cannot. This is not a communication failure — it is a structural consequence of the learning process through which expertise is acquired. The Dreyfus framework confirms that no amount of improved exit interviewing will access this knowledge, and that the only viable capture approach observes the expert's behavior in the environment rather than asking the expert to describe it.

A critical refinement is required. Peña (2010), in a pointed critique of the Dreyfus model, argued that even expert practitioners do not operate exclusively in intuitive mode but rather alternate between intuitive and analytical modes depending on situational demands. Peña's objection is empirically well-supported — protocol analysis studies consistently show experts shifting into explicit reasoning when confronted with atypical cases. We accept this refinement and note that it is accommodated *within* the CIAER schema rather than against it. Klein's three RPD variants (Section 2.9) are precisely the topological variants of Intuition that correspond to Peña's mode-alternation: Variation 1 (Simple Match) captures pure intuitive recognition; Variation 2 (Situation Diagnosis) captures the rule-based mode; Variation 3 (Action Evaluation) captures the deliberate analytical mode with mental simulation and candidate rejection. Peña's critique demonstrates that the Dreyfus model in its strongest form is too strong, not that expertise capture is impossible;

the capture architecture must accommodate all three modes, which CIAER does through SRK-level tagging and RPD-variant classification of each event.

2.8 Rasmussen's SRK Hierarchy: Capture Intensity by Cognitive Mode

Jens Rasmussen's SRK framework (1983) identifies three fundamentally distinct cognitive modes through which expert behavior is executed. Every CIAER event record carries an SRK classification tag that determines the appropriate capture intensity and downstream modeling approach.

Skill-Based Behavior: Automatic, procedural execution without conscious attention. Operates below the threshold of articulable consciousness per Dreyfus and Ericsson and Simon. Captured through action observation, biometric somatic signals, and gaze engagement data alone. Voice annotation is neither expected nor reliable for this class.

Rule-Based Behavior: Pattern matching against stored experiential rules. Recognition triggers a memorized response. Minimal contextual annotation is sufficient.

Knowledge-Based Behavior: Deliberate analysis and mental simulation in novel situations requiring model revision. The highest-value capture target where the full multimodal stack provides maximum yield.

2.9 Klein's Recognition-Primed Decision Model

Gary Klein's RPD model (Klein, Orasanu, and Calderwood, 1993) provides the structural account of how expert intuitive choice operates, elaborating the Intuition phase of the CIAER schema across three topological variants: *Variation 1 — Simple Match* (Skill-based), where a highly typical situation is recognized and the CIAER graph collapses toward a near-direct Cause-Action arc; *Variation 2 — Situation Diagnosis* (Rule-based), where an atypical situation requires active information gathering and the Intuition node expands to include iterative loops back to Cause; and *Variation 3 — Action Evaluation* (Knowledge-based), where multiple candidate actions are serially evaluated through mental simulation with full projection modeling and rejection loops.

2.10 Endsley's Situation Awareness: The Projection Dimension

Mica Endsley's Situation Awareness model (1995; see also Endsley and Jones, 2012) identifies three hierarchical levels: Level 1 Perception (mapping to Cause), Level 2 Comprehension (mapping to Intuition), and Level 3 Projection — the anticipation of future system states based on current trajectory. Level 3 Projection is the most valuable and least-captured dimension of tacit expertise: the capacity to intervene before threshold violation rather than after. It is the forward prediction function of the generative model operating at a longer timescale than immediate perception, and it is entirely invisible to procedure-capture platforms that document actions after they occur.

2.11 Boyd's OODA Loop Revisited: Orientation as the Model Archive

Boyd's Orientation phase — described as shaped by “genetic heritage, cultural tradition, previous experiences, and analytical ability” — is the OODA Loop's closest equivalent to what CIAER calls the generative model. Boyd recognized that the quality and speed of Orientation, not the speed of Observation or Action, determines decision superiority. This insight maps directly onto the CIAER hypothesis: the depth and accuracy of the accumulated bidirectional reconciliation history encoded in the generative model is what distinguishes the expert from the competent practitioner.

2.12 Kolb's Experiential Learning Cycle

David Kolb's Experiential Learning Cycle (1984) describes a four-phase cycle — concrete experience, reflective observation, abstract conceptualization, active experimentation — developed within organizational learning theory to explain how practitioners build knowledge from experience. The cycle directly maps onto CIAER's structure (Experience = Cause + Action + Effect; Reflection = Result; Conceptualization = updated Intuition model; Experimentation = the next Action). Kolb explicitly addressed the organizational knowledge transfer problem that motivates this paper: learning from experience cannot be replaced by explicit instruction. The correspondence between Kolb's cycle and CIAER — developed from entirely different intellectual traditions — constitutes an additional convergent data point for the bidirectional schema hypothesis.

2.13 Friston's Free Energy Principle and Active Inference

Karl Friston's Free Energy Principle (FEP) and Active Inference framework provide the most mathematically rigorous theoretical grounding for the CIAER bidirectional schema. Before presenting the mapping, we must acknowledge several important characteristics of the FEP. First, it is a variational principle expressed in the mathematics of Bayesian mechanics, Markov blankets, and variational free energy minimization. As Friston himself has emphasized, the FEP is a mathematical principle rather than a falsifiable empirical theory — in this respect it is analogous to the principle of least action or the principle of maximum entropy. The empirical scope and falsifiability of the FEP itself remain subjects of active philosophical debate (Colombo and Wright, 2017; Bruineberg et al., 2022). Second, the empirical status of predictive coding, the broader neuroscientific family to which active inference belongs, has been characterized in recent reviews as supported by converging but still modest evidence — strong in specific regions of visual cortex, weaker elsewhere in the brain. The mapping between CIAER and the FEP presented below is therefore a functional isomorphism at the level of behavioral description, not a claim of mathematical equivalence, formal derivation, or neuroscientific settledness. Our use of the FEP as a theoretical lens does not depend on the resolution of these debates, but we flag them so that the reader knows precisely what the mapping is and is not claiming.

Under the FEP, any self-organizing system that persists across time minimizes the discrepancy between its internal generative model and incoming sensory data. The behavioral mapping onto CIAER is as follows: Cause corresponds to the arrival of a sensory prediction error. Intuition corresponds to perceptual inference — updating the generative

model to identify the most probable cause of the prediction error. Action corresponds to active inference — the agent acts to bring about the sensory states its model predicts. Effect is the new sensory state generated by the action. Result is the free energy minimization cycle's completion, either confirming the model or generating a new prediction error initiating the next cycle.

2.14 Damasio's Somatic Marker Hypothesis: Neurobiological Grounding of the Biometric Layer

Antonio Damasio's Somatic Marker Hypothesis (SMH; *Descartes' Error*, 1994) provides the theoretical grounding for the biometric integration architecture. Damasio demonstrated through lesion studies that individuals who retain intact reasoning capacity but lose access to somatic responses to prior outcomes make systematically poor decisions under uncertainty. His conclusion was that the body's physiological responses — EDA changes, HRV shifts, skin temperature changes — encode the accumulated outcome history of prior decision cycles and bias cognition toward advantageous actions before deliberate analysis begins.

Two qualifications are essential. First, Damasio's original experimental paradigm involved human participants making economic decisions under uncertainty in a controlled laboratory setting — the Iowa Gambling Task. The application of the somatic marker hypothesis to industrial expert judgment in operational environments is a theoretical extension that, while consistent with Damasio's framework, requires its own empirical validation. We advance this extension as a theoretically grounded hypothesis rather than an established fact. Second, as Overskeid (2021) and other recent critics have noted, the SMH has accumulated a range of stronger interpretations that outrun what Damasio's own evidence supports. Our use of the SMH is deliberately restricted to the claim that Damasio himself defends: that somatic signals correlate with and bias decision-making under uncertainty in ways detectable through physiological measurement. We do not claim that somatic markers uniquely determine decisions, that they are the sole or primary mechanism of expert judgment, or that the biometric layer accesses the content of expert cognition directly. It accesses a physiological correlate of decision engagement, which is useful precisely because Dreyfus-level expert cognition is otherwise inarticulable.

2.15 Convergence Summary

The convergence documented above operates at two levels. At the coarse level, all eleven frameworks describe a closed feedback cycle between agent and environment — a finding that is unsurprising and does not by itself support CIAER's specific decomposition. At the fine level, the frameworks converge on a five-component structure that distinguishes environmental detection from model inference, model inference from action execution, action execution from environmental response, and environmental response from model update. It is this fine-grained convergence, not merely the shared cyclic structure, that supports the CIAER hypothesis.

The strength of the convergence argument derives from cross-disciplinary agreement rather than full statistical independence between frameworks. Several of these traditions

have influenced one another: Klein, Endsley, and Rasmussen emerged from overlapping NDM and human factors communities; Dewey, cybernetics, and the FEP share an intellectual lineage through predictive processing; Kolb explicitly draws on Dewey. These co-evolutions are acknowledged and do not diminish the convergence argument — they indicate that multiple researchers working within broadly related empirical traditions have arrived at structurally equivalent descriptions of the agent-environment interaction cycle.

An instructive parallel to this convergence argument appears in recent comparative linguistics. Verkerk et al. (2026), analyzing the Grambank database of grammatical features across over 2,000 languages using Bayesian spatiophylogenetic methods that control for both shared ancestry and geographic proximity, found that approximately one-third of proposed linguistic universals survive rigorous statistical testing — and that languages actively converge toward these preferred configurations over evolutionary time, with rates of change toward “harmonic” states significantly exceeding rates toward disharmonic states. Their strongest results concerned hierarchical implicational universals: structural constraints where the presence of a rarer feature implies the presence of a more common one within the same paradigm. This finding is directly relevant to CIAER: just as shared cognitive and communicative pressures push the world’s languages toward the same structural solutions despite enormous combinatorial freedom, we hypothesize that shared cognitive and environmental pressures push expert decision-making in dynamic physical environments toward structurally convergent solutions describable by the CIAER schema. The Verkerk et al. methodology — testing proposed universals against null hypotheses while controlling for non-independence, then separately testing for co-evolutionary dynamics — provides a template for future large-scale validation of CIAER patterns across facilities, industries, and operator lineages.

2.16 Causal Confusion and the Pathology of Behavioral Cloning

A final convergent signal for the CIAER thesis arrives from an unexpected direction: the machine learning literature on imitation learning. De Haan, Jayaraman, and Levine (2019) identified a counterintuitive failure mode they termed *causal confusion*. A policy trained on expert demonstrations through behavioral cloning learns to predict expert actions from observations; however, the trained model cannot distinguish observations that *cause* expert actions from observations that are merely *correlated* with expert actions. Their canonical illustration is an autonomous driving policy that learns to brake when the brake indicator is illuminated — because in the expert demonstrations, the brake indicator and the braking action are perfectly correlated. The model has learned a non-causal proxy (effect) for the causal signal (the pedestrian ahead). Access to more information, paradoxically, makes this worse: the richer the observation space, the more spurious correlates are available for the policy to latch onto.

Ortega et al. (2021) extended this result to large sequence models, showing that the same pathology arises when transformers are trained to imitate expert behavior as next-token prediction. Their diagnosis is that conditioning on past actions treats actions as ordinary observations, when they should be treated as *causal interventions* — which means the demonstration data contains a latent causal structure that behavioral cloning systematically fails to recover. Ruan et al. (2023, 2024) formalized the problem further,

showing that exact imitation is impossible under unobserved confounding without additional causal assumptions, and developed robust algorithms for the partially identifiable regime. Shao et al. (2024) extended the instrumental-variables toolkit to the problem.

The implication for industrial expertise capture is stark. A veteran operator's behavior, captured as a stream of sensor readings and action logs, contains all the information the operator observed *and* all the actions taken — but behavioral cloning on this stream will systematically confuse the operator's causes-for-action with correlates-of-action. The operator who reduces screw speed on hearing a subtle auditory change plus observing a pressure fluctuation has a specific causal model: auditory change + pressure drift $\Rightarrow$ melt viscosity drift $\Rightarrow$ reduce screw speed. A pure behavioral cloner may instead learn: *recent pressure fluctuation**** ***** $\Rightarrow$ ***** *****reduce screw speed, *or worse*, time-of-shift pattern**** ***** $\Rightarrow$ ***** *****reduce screw speed*, because these spurious correlates are available and do not require the cloner to recover the underlying causal structure the operator is actually using.

This is precisely the problem the CIAER decomposition is designed to address. By structuring each captured event into *Cause* (the specific prediction error that initiated action), *Intuition* (the internal inference, voice-annotated where articulable and biometrically-signed where not), *Action* (the intervention taken), *Effect* (the environmental response), and *Result* (the model update registering confirmation or disconfirmation), CIAER supplies the causal skeleton that behavioral cloning otherwise has to reconstruct — and typically fails to reconstruct — from surface behavior alone. The five-phase structure *is* the causal DAG of the decision event, recorded by construction rather than inferred post hoc. The externalization bottleneck (Section 2.6) and the causal confusion bottleneck are, we argue, the same bottleneck viewed from two disciplines: both describe the failure of information about *why* an expert acted to transmit alongside information about *what* the expert did.

This convergence is theoretically consequential and practically useful. It provides the strongest technical validation of the framework's central thesis — that capture of causal structure, not merely behavioral surface, is what distinguishes an expertise-preserving corpus from a procedural log. It also supplies the most specific engineering requirement the framework imposes: that each captured event must be annotated with its cause, not merely with its action, or the captured corpus will inherit the causal-confusion pathology of ordinary behavioral cloning. A downstream argument, which we develop in Section 6.4, is that the ArcShield corpus is accordingly designed to make the causal structure extractable by construction, and that Hu and Clune's Thought Cloning paradigm — which augments behavioral cloning with contemporaneous reasoning traces — provides the natural training objective for models learned from such a corpus.

2.17 Pupillometric Evidence for Anticipatory Inference

The preceding convergence argument is structural: eleven frameworks plus the causal-confusion literature agree on a shared behavioral decomposition. A separate line of

evidence, drawn from cognitive neuroscience, speaks to the *mechanism* of the Intuition phase in particular.

Since the foundational work of Hess and Polt (1964) and the systematic research program established by Kahneman (1973), pupil dilation has been recognized as a precise real-time correlate of cognitive effort during mental arithmetic. What has changed recently is the interpretation. Zénon, Salvaggio, and Andres (2026), in a peer-reviewed study of arithmetic problem-solving, measured pupil dilation across the interval between presentation of a first operand and a second operand in a sum. The pupil began to dilate at the first operand — long before the problem was fully specified — and the magnitude of that early dilation was proportional to the information gain the first operand contributed, that is, how much it narrowed the prior distribution over plausible answers. Larger early dilation also predicted faster response times once the full problem arrived. The brain was updating a posterior over the answer before the problem had finished being stated.

The implication for CIAER’s Intuition phase is twofold. First, it supplies a neurophysiological correlate for the claim that forward projection — inference over a not-yet-specified terminal state — is a native feature of cognitive processing rather than a late-stage deliberative add-on. This sits directly alongside Endsley’s Level 3 Projection (§2.10) and Klein’s mental-simulation variant of RPD (§2.9), and supplies a formal vocabulary — sequential Bayesian updating of a posterior over outcomes — for describing what those frameworks name at a higher level of abstraction. Second, it demonstrates that this updating process is *involuntarily* detectable through a biometric channel (pupillometry) even when no overt behavior is produced. This reinforces the architectural claim of Section 5.2: the biometric integration layer accesses decision processing that voice annotation and action logs cannot reach, because the Intuition-phase computation begins before the expert has anything articulable to report and remains partially inaccessible to introspection even afterward.

One qualification warrants explicit statement. The Zénon paradigm is closed-world: participants know in advance the range of possible answers, so “Bayesian updating” has a well-defined prior distribution. Operational expertise in the domains CIAER targets is open-world in comparison — experts do not carry explicit probability distributions over outcomes, and their priors are implicit in the accumulated reconciliation history of the generative model. The closed-world result therefore establishes the mechanism’s existence cleanly but does not by itself prove its full operational generality. The argumentative move is conservative: if anticipatory Bayesian updating is robustly detected in constrained arithmetic, where priors are shallow and stakes are negligible, its presence in operational contexts — where priors are built from decades of reconciliation history and uncertainty reduction is consequential — is a plausible extension of the same mechanism rather than a leap to a qualitatively different process. Direct pupillometric validation in operational contexts is identified as a research aspiration in Section 10.9 alongside the EEG program.

3. The CIAER Schema: Formal Definition of the Bidirectional Loop

The CIAER schema is formally defined as a directed cyclic graph whose five nodes represent the five phases of the bidirectional reconciliation loop and whose edges represent both the sequential flow of each cycle and the recursive feedback connecting cycle completion to cycle initiation. The schema describes the proposed minimum complete structure of adaptive purposeful behavior at the behavioral level of analysis — the level at which actions, environmental states, and their consequences are observable — without prescribing the internal architecture through which any specific system implements those phases.

3.1 The Five Phases

Cause — The Interface Event: A registered discrepancy between the internal model's current prediction and the actual environmental state. Cause is simultaneously the terminal point of the inward arc and the initiating point of the outward arc. Captured via camera, machine telemetry, and biometric somatic signals; gaze dwell marks the precise moment of visual engagement with the causative element.

Intuition — Inference and Projection: Internal model update to identify the most probable cause of the prediction error, evaluate candidate actions, and project probable system trajectory. Encompasses the SRK-level-dependent depth of processing from immediate pattern retrieval (Skill-based) through stored rule application (Rule-based) to deliberate mental simulation (Knowledge-based). Physiological signatures associated with this phase — including HR elevation, HRV suppression, pupil dilation, and EDA changes — are captured where the deployed hardware allows. The current operational trigger uses HR, HRV-RMSSD, and motion-gated gaze dwell; EDA and pupillometry are research aspirations (Section 5.2). Pupillometric evidence (Zénon et al., 2026; §2.17) indicates that this phase begins updating a posterior over outcomes before all inputs have been observed, consistent with anticipatory Bayesian inference rather than a discrete post-observation computation.

Action — Outward Arc Execution: The agent executes the selected intervention, actively restructuring the external world. Action nodes may expand into hierarchical sub-graphs of nested micro-CIAER cycles, reflecting the hierarchical structure of the generative model.

Effect — Prediction Test: The external world responds to the agent's intervention, generating a new environmental state that empirically tests the model's prediction.

Result — Model Update and Loop Closure: The agent evaluates the Effect against the model's prediction. A confirming Result reinforces the encoded reconciliation pathway. A disconfirming Result generates a new prediction error and initiates model revision. Result feeds directly back to Cause as the environmental condition initiating the next cycle.

3.2 The Bidirectional Loop

The bidirectional structure operates as two simultaneously active arcs sharing the Cause node as their interface point. *Outward Arc:* Cause → Intuition → Action → Effect → Result.

Inward Arc: Result → Cause (next cycle) (model update is internal to Result). *Interface:* Cause marks the boundary between internal model and external world.

The loop is closed and recursive. There is no natural terminus within a given operational engagement; it runs continuously as long as the agent maintains active sensorimotor contact with the environment. The interval between cycles varies enormously — milliseconds in Skill-based reflex responses to hours in complex Knowledge-based deliberation — but the structural architecture of each cycle is invariant. This invariance is the CIAER hypothesis's testable core: if the schema fails to describe the behavioral structure of expert action in any domain where purposeful action occurs, the hypothesis requires revision.

3.3 What CIAER Adds Beyond Prior Frameworks

The relationship between CIAER and the eleven convergent frameworks is one of synthesis and specification, not replacement. CIAER contributes the following elements not present in any single prior framework: the explicit Effect phase as a distinct, sensor-captured environmental event; the explicit Result phase with bidirectional model update capturing disconfirmation events; SRK-level classification determining appropriate capture intensity per event class; the physiological capture architecture bridging the Dreyfus inarticulability problem; multi-expert divergence handling preserving parallel expert subgraphs; and the operational knowledge management application specifying an architecture for systematic capture, structuring, and preservation of the bidirectional reconciliation history.

3.4 Multi-Expert Divergence as Epistemic Signal

When two expert agents observe the same Cause event and produce different Intuition-to-Action paths, this reflects the operation of two distinct generative models built from different reconciliation histories responding to the same prediction error. The CIAER knowledge graph preserves these divergent chains as parallel subgraphs rather than collapsing them into consensus — because we hypothesize that the divergence points, the precise environmental conditions under which two expert models produce different action recommendations, are among the highest information-density nodes in the knowledge graph, revealing latent variables that distinguish master-level from expert-level model architecture. Empirical validation of this claim is designed for the filter assembly department described in Section 7, where multiple operators performing identical tasks with measurable downstream quality outcomes will enable direct correlation between divergent chain structures and output quality.

3.5 Hierarchical Nesting

Expert decision-making in complex physical environments is rarely atomic. A macro-level intervention contains dozens of micro-CIAER cycles within the Action phase. The schema accommodates this through hierarchical nesting: any Action node can expand into a child CIAER sub-graph, and any Cause node can simultaneously trigger parallel CIAER chains. The CIAER graph hierarchy mirrors the hierarchical structure of the agent's generative model itself.

4. Cross-Domain Manifestations of the Bidirectional Reconciliation Loop

Having defined the CIAER schema formally, we now examine whether its predicted structure manifests across operationally diverse domains. The following cross-domain analysis presents six operational instances consistent with the CIAER schema, drawn from domains that differ substantially in sensory modalities, decision timescales, and — critically — the physical vs. abstract character of the reconciliation space. The first five instances describe experts operating in physically embodied processes where prediction errors register through direct sensory contact with machinery or material. The sixth instance describes an expert operating in a domain whose “environment” is an abstract dashboard of ratio-based performance metrics rather than a physical process — a case that tests whether the CIAER schema’s structural invariance extends beyond physically embodied domains into the management of purely informational systems. Each domain’s practitioners would be operating the same minimum sufficient behavioral schema through different environmental vocabularies; the schema structure would be invariant while the sensory inputs, action repertoires, and accumulated reconciliation histories differ. These cross-domain instances are described as *consistent with*, rather than *proof of*, the hypothesis.

4.1 Polymer Processing: PVC Extrusion

Cause: Visual discoloration toward yellow in the extrudate combined with fractional variance in pressure transducer readings — a multimodal prediction error against the expected process state.

Intuition: Model inference identifies excessive frictional heat generation as the most probable causal state; somatic markers retrieve accumulated outcome history for this event class; Level 3 Projection models degradation trajectory if untreated.

Action: Reduction of screw speed via variable speed motor control, or adjustment of cooling water flow to specific barrel zones.

Effect: Reduced mechanical shear lowers internal melt temperature; cooling adjustments stabilize pressure dynamics.

Result: Extrudate returns to clear, water-white profile; gauge readings confirm dimensional tolerances. Prediction confirmed; model reinforced.

4.2 Healthcare: Surgical Intervention

Cause: Subtle change in tissue texture and color of surrounding vascular structures, concurrent with anomalous hemodynamic telemetry.

Intuition: Inference identifies probable concealed micro-hemorrhage; somatic urgency profile retrieved from accumulated surgical history.

Action: Adjusted laparoscope angle, altered dissection trajectory, targeted hemostatic tool deployment.

Effect: Hemostatic intervention arrests localized bleeding.

Result: Patient vital signs normalize; surgical field clarity restored. Outcome integrated into generative model.

4.3 Industrial Fabrication: Advanced Welding

Cause: Anomalous auditory frequency — erratic, sputtering arc sound — combined with a turbulent, excessively wide weld puddle.

Intuition: Model identifies excessive heat input and insufficient travel speed; somatic markers retrieve tactile and acoustic reconciliation history.

Action: Increased torch travel speed and micro-adjustments to Contact-Tip-to-Work Distance.

Effect: Altered thermal dynamics of the melt pool modify filler metal flow and penetration depth.

Result: Weld puddle stabilizes; flat, consistent bead profile with optimal structural integrity produced.

4.4 Aviation: Pilot Command and Control

Cause: Marginal discrepancy between central attitude indicator and supporting instruments, with concurrent change in engine acoustics.

Intuition: Recognition of incipient stall signature from accumulated flight-hour reconciliation history; Level 3 Projection of altitude loss trajectory.

Action: Pitch and bank correction while advancing throttle.

Effect: Movement of control surfaces alters aerodynamic profile; increased thrust arrests altitude loss.

Result: Instrument scan confirms restoration of level flight within safe parameters.

4.5 Precision Agriculture: Agronomic Management

Cause: Near-infrared reflectance anomalies in multi-spectral drone imagery cross-referenced with low soil moisture telemetry.

Intuition: Holistic model inference distinguishes localized water stress from systemic disease, drawing on accumulated seasonal history.

Action: Variable Rate Technology programs center-pivot irrigation system to alter transit speed and water application rate.

Effect: Targeted delivery of water and micro-nutrients alters soil hydrology and biochemical environment.

Result: Follow-up sensor readings confirm restoration of crop vigor. Intervention validated.

4.6 Service Operations: Fast-Food Restaurant Shift Management

The preceding five instances all describe experts embedded in physically embodied processes — their “environment” is machinery, tissue, filler metal, an aircraft, or a crop field, and the reconciliation loop is mediated by direct sensory contact with physical material. We include a sixth instance drawn from a domain whose “environment” is abstract — a dashboard of ratio-based performance metrics — precisely because it tests whether the CIAER structural invariance extends beyond physical embodiment. If the schema holds in a service-operations setting where the operator acts on a display rather than on matter, the domain-agnostic structural invariance claim is strengthened; if it fails, the minimum-sufficient-schema claim requires qualification to the physical-process domain.

Cause: A shift manager monitors multiple interdependent operational metrics visible on a back-office dashboard: labor percentage trending toward a threshold ceiling for the current day-part, sales per man-hour running below target, average order-completion time drifting above the corporate standard, and scheduled staffing levels for the next time slot compared against projected traffic. The Cause registers when the manager perceives a gap between the current trending numbers and the required threshold values that must be achieved before the current time slot closes — a prediction error against the expected trajectory of the operational envelope. Gaze dwell on the specific dashboard tile registering the largest deviation marks the capture moment.

Intuition: The manager performs a pattern-matched forward projection of the trajectory under time pressure, integrating multiple correlated streams: if labor percentage continues rising at the current rate and sales per man-hour continues falling at the current rate, the day-part closes outside tolerance and a corrective intervention window is narrowing. The inference is not an explicit calculation; the manager does not reach for a calculator or open a spreadsheet. The Intuition phase operates as a Recognition-Primed Decision Variation 2 or 3 — depending on situation atypicality — drawing on accumulated shift-management history to project where the numbers will land if no intervention is made and to identify which levers are available within the remaining time window. A somatic signature (mild stress response, increased task focus) accompanies the projection when the trajectory is perceived as exceeding comfortable margins.

Action: The manager selects from a constrained vocabulary of discrete staffing interventions: send a specific employee on an unscheduled break to reduce active labor cost during a slower interval; cut an employee early from the shift to reduce labor hours booked against the day-part; hold an employee past scheduled end-time to prevent an anticipated understaffing gap; re-route a scheduled employee from front-counter to drive-thru based on a projected traffic mix shift; or hold staffing steady and accept

short-term margin compression in exchange for service-level protection. Each intervention is discrete, low-latency, and reversible within the same day-part.

Effect: The dashboard's measurable response to the intervention over the remaining time window of the day-part. Labor percentage adjusts downward as hours stop accumulating; sales per man-hour adjusts as the labor denominator changes; order-completion time responds to the new staffing configuration.

Result: The manager evaluates the day-part close against the required thresholds. A confirming Result — all metrics within tolerance — reinforces the pattern-match: *this trajectory at this clock time, with this staffing configuration, corrected by this intervention*. A disconfirming Result — a metric closed out of tolerance despite the intervention — updates the generative model with a specific lesson about where the manager's projection was wrong: the intervention was too late, the lever chosen was weaker than expected, the underlying traffic trajectory was mis-read. The updated model feeds directly into the next day-part's Cause-detection.

The structural point is that the CIAER loop operates here with identical topology to the PVC extrusion case, despite no physical process being operated on. The Cause is an abstract prediction error against a multi-channel ratio dashboard rather than a visual-plus-pressure cue; the Intuition is forward-projection of a multi-variable trajectory under time pressure rather than a thermodynamic inference; the Action is a staffing intervention selected from a constrained operational vocabulary rather than a mechanical adjustment; the Effect is a dashboard response rather than a melt-behavior response; the Result is a threshold-conformance evaluation rather than a product-quality evaluation. The surface vocabulary is entirely different. The underlying schema is invariant. This is the prediction the CIAER hypothesis makes for domain-agnostic structural invariance, and the service-operations instance — because it operates on ratios rather than matter — constitutes a more stringent test of that invariance than another physical-process instance would.

The Intuition phase in this domain also surfaces the anticipatory-inference mechanism described in §2.17 with unusual clarity. The shift manager's forward projection over the day-part's terminal state is not a calculation performed at the moment of intervention — it is a continuously updated posterior, revised with each new observation (an early cadence spike, a staffing no-show, a drive-thru slowdown, a front-counter queue beginning to form) long before the day-part has closed. Intervention fires when that posterior crosses a risk threshold, not when the arithmetic over labor percentage is "complete." Zénon et al.'s finding that pupillary posterior-updating over arithmetic outcomes begins at the first operand, before the problem is fully specified, is a laboratory-scale correlate of exactly this operational pattern translated to a shift-management timescale. The externalization pathway for such expertise therefore cannot be retrospective reconstruction; whatever channel captures it must run *during* the updating, as CIAER's biometric and dashboard-interaction streams are designed to.

A secondary point from this instance concerns the *externalization bottleneck*. A veteran shift manager of this kind tracks four to seven interdependent ratios simultaneously,

performs real-time multi-variable projection against each, and selects staffing interventions from a discrete vocabulary with sub-second latency — all without explicit calculation and typically without the ability to articulate the projection logic when asked. The conventional documentation response — a shift-management SOP — captures none of this; it captures rules for when to hire and when to terminate, not the continuous day-part-level reconciliation that separates a high-performing shift manager from an average one. The CIAER capture architecture applied to this domain would use dashboard-interaction capture as the Cause and Action sensor stream, biometric signals as the Intuition sensor, and the day-part close as the Result sensor — with voice annotation reserved for the occasional Knowledge-based event where the manager encounters a novel situation requiring deliberate analysis. This is the CIAER externalization pathway applied to an abstract-environment domain, and it is invisible to any procedure-capture platform focused on physical operational steps.

5. The Hardware Strategy: AI Glasses as a Data Platform

To capture the CIAER bidirectional loop as it occurs, organizations require a continuous, uninterrupted stream of localized, first-person data encompassing both arcs. The strategic solution articulated in this paper is to leverage commodity AI glasses as the primary point-of-view capture device, paired with continuous-monitoring wrist wearables for the biometric layer, and to integrate capability into equipment that workers already accept as comfortable and unobtrusive. This section describes the target hardware architecture toward which ArcShield is converging; Section 9 describes the current proof-of-concept implementation on a mounted smartphone platform, which is an interim deployment mode that has been sufficient to validate the pipeline.

5.1 AI Glasses as the Point-of-View Sensor Platform

Meta released the Wearables Device Access Toolkit in public developer preview in December 2025, providing the first commercial SDK that allows third-party mobile applications to access the camera and audio streams of a mass-market AI glasses product (Ray-Ban Meta Gen 1, Gen 2, and Ray-Ban Meta Display) through iOS and Android SDK libraries. The architecture is specifically suited to the CIAER capture problem: the glasses stream their first-person camera frames, microphone audio, and speaker output to a paired mobile device via the Bluetooth and Wi-Fi profiles the SDK exposes, and the paired device runs the application logic. This matches the ArcShield computational architecture directly — the app runs on the operator's phone, where the fusion processing already lives — and requires no custom hardware development.

The functional fit is strong. First-person camera capture from an operator's actual line-of-sight records the exact environmental Cause that triggered the decision event, eliminating the framing and pointing artifacts that mounted or handheld cameras introduce. The glasses' open-ear speakers and microphones provide a voice annotation channel that does not interfere with hearing protection compliance and does not require the operator to remove or manipulate a separate device. Hands-free operation is preserved throughout — a critical constraint in manufacturing environments where operator

attention, both hands, and peripheral vision must remain fully available to the task. The developer preview currently exposes POV photo and video capture, frame streaming for real-time processing (object detection, OCR, scene understanding from multimodal LLMs), and programmatic capture triggers from app logic, voice commands, or contextual events. General availability for broad third-party publishing is targeted for 2026.

Two caveats apply. First, the developer preview does not expose Meta AI “Hey Meta” voice invocations to third-party apps; the ArcShield voice annotation channel uses the platform’s standard microphone streaming rather than the Meta AI capability. Second, the SDK’s policy frame restricts sensitive use cases, and enterprise deployment paths will require engagement with Meta’s partner program as the ecosystem matures. These are manageable constraints; neither blocks the core capture architecture.

5.2 The Biometric Integration Layer: Accessing the Somatic Archive

The biometric integration layer is grounded in the Somatic Marker Hypothesis as a theoretical extension to the industrial context, as qualified in Section 2.14. By establishing personal baselines during routine shift operations for continuously-monitored physiological channels, the system detects statistically significant deviations that, under this hypothesis, correspond to arousal events associated with expert decision engagement — automatically initiating the CIAER capture sequence without requiring conscious operator action. This passive trigger architecture directly addresses the Dreyfus inarticulability problem and its Peña refinement: the somatic signal fires whether or not the expert consciously registers a decision and independently of whether the expert is currently in intuitive or analytical mode.

A distinction between theoretical grounding and operational implementation is required. Damasio’s framework most directly predicts that electrodermal activity (EDA), mediated by the sympathetic branch of the autonomic nervous system, would carry the richest somatic-marker signal. Continuous wrist EDA (cEDA) is physically measured by the Fitbit Sense 2 and Pixel Watch 2/3/4 sensor array, but the raw cEDA waveform is not exposed to third-party developers on current Google platforms — it is consumed internally by Fitbit’s Body Response stress-detection feature and surfaced only as a derived notification. Raw cEDA access is therefore treated in this paper as a research aspiration alongside direct operational pupillometry and EEG (Sections 10.5 and 10.9), not as a deployed trigger channel.

The operational trigger at the current deployment tier is accordingly constructed from the channels that *are* exposed to third-party developers through the Wear OS Health Services and Health Connect APIs on the Google Pixel Watch 4:

- **Continuous optical heart rate** — all-day photoplethysmography with on-device machine-learning noise rejection. Heart rate functions both as an independent arousal channel and as the basis for heart rate variability computation. HR has been shown to discriminate cognitive load conditions with high accuracy in large-N operational studies, including a study of 892 drivers in which HR achieved

within-subjects accuracy of 0.94 discriminating 30-second windows of differing cognitive load (Hopstaken et al., 2024).

- **Heart rate variability (HRV, RMSSD)** — continuously computed by Fitbit's on-device algorithms and exposed to developers through the Health Connect HeartRateVariabilityRmssdRecord schema. RMSSD is the most widely used time-domain HRV index in psychophysiological research because it reflects rapid phasic vagal modulation of the sinoatrial node and is statistically well-behaved relative to alternative indices. A systematic review of pilot cognitive workload studies (Ngo et al., 2024, *Sensors*) and a direct-comparison study of wearables against multi-lead ECG during mental workload tasks (Lo et al., *Journal of Cognitive Engineering and Decision Making*, 2017) support RMSSD as a validated operational marker of cognitive load during non-physically-demanding supervisory activity. A pilot study in an emergency-department setting (Rose et al., 2025) demonstrated that a small biometric feature set combining HR, RMSSD, LF/HF ratio, and skin conductance discriminated high subjective cognitive load across 864 fifteen-minute intervals during live clinical shifts — providing direct evidence that the HRV-based arousal-detection approach works under the noisy, motion-rich conditions characteristic of operational environments.
- **Accelerometer and gyroscope streams** — continuous six-axis motion data used on-device for activity classification. The activity classification serves a critical confound-mitigation role: it allows the system to gate arousal-event detection to periods of relatively stationary operator activity, where HR and HRV deviations are more plausibly attributable to cognitive rather than physical sources.
- **SpO2 and loss-of-pulse detection** — continuous safety-oriented channels available for corroborative signal fusion and for monitoring operator physiological status independently of the decision-event pipeline.
- **Fitbit Body Response** — the derived stress notification, which fuses cEDA, HR, HRV, and temperature signatures in Fitbit's proprietary algorithm and surfaces to third parties as an event stream. Body Response is retrospective and aggregated rather than a real-time waveform, but it is consumed as a corroborative signal: a Body Response notification firing in the same window as a multi-channel HRV/HR/gaze-dwell deviation raises the confidence of the captured event.

The resulting trigger architecture is multi-signal by construction. The primary channels are HRV-RMSSD deviation from personal baseline and HR deviation from personal baseline, gated on accelerometer-derived activity classification (stationary or low-motion operator state), and corroborated by gaze-dwell engagement captured by the glasses layer (Section 5.1). A capture event fires when two or more of these channels register convergent deviation within a calibration window; single-channel deviations are logged but do not trigger capture. This multi-signal requirement is a confound-mitigation response, not just a robustness preference — it reduces the probability that a transient cardiovascular event (caffeine spike, social interaction, physical exertion leaking past the activity classifier) produces a false capture.

One important limitation of the current Pixel Watch 4 platform warrants explicit statement: skin temperature is not a continuously-sampled waking-hours channel. The skin-temperature-variation-from-baseline feature samples during sleep and reports nightly variation; it is not available as a real-time shift-operations signal. Skin temperature therefore contributes to the biometric layer only indirectly through the Fitbit-computed Body Response output, which incorporates temperature in its proprietary fusion.

5.3 Developer SDK Stack

The CIAER capture architecture integrates three complementary developer SDKs, each handling one functional layer:

Meta Wearables Device Access Toolkit (iOS/Android; developer.meta.com/wearables) handles the point-of-view camera and microphone access for the glasses. Frames and audio stream to the paired phone where the ArcShield application runs.

Wear OS Health Services API ([androidx.health.services.client](https://developer.android.com/health-services)) handles on-watch sensor access for continuous channels. The API exposes three clients: MeasureClient for real-time spot-rate updates suitable for short-lived foreground UI; ExerciseClient for continuous workout-mode streams with device-native fusion; and PassiveMonitoringClient for power-efficient background monitoring suitable for all-shift sampling. The Health Services API is the supported Google path for accessing Pixel Watch 4 sensor streams with on-device fusion and power management already handled by the platform.

Health Connect API ([androidx.health.connect.client](https://developer.android.com/health-connect)) handles on-phone aggregation of normalized biometric streams across wearable vendors. Health Connect defines a vendor-neutral schema — HeartRateRecord, HeartRateVariabilityRmssdRecord, SkinTemperatureRecord, ActiveCaloriesBurnedRecord, ExerciseSessionRecord, and related types — which allows the application layer to consume biometric data through a stable interface while the source device can be a Pixel Watch, Samsung Galaxy Watch, Garmin device, or any other Health-Connect-compatible wearable. Health Connect is the recommended Google path for the Sensor Abstraction Layer of Section 6.1.

A deliberate architectural consequence of the vendor-neutral Health Connect abstraction is that the trigger stack is not locked to Pixel Watch hardware. For deployments or validation phases that require access to channels not exposed by the consumer-wearable platform — particularly raw-waveform EDA for research-grade ground-truth collection — alternative wearables such as the Empatica EmbracePlus (medical-grade EDA, HR, skin temperature, accelerometer at research sampling rates) can be substituted while the production application layer remains unchanged. This is discussed further in Section 7 as part of the multi-domain validation program.

5.4 Known Biometric Confound Sources and Mitigation Strategy

The HRV/HR-based arousal trigger faces several known confound sources in industrial environments that must be acknowledged and addressed. Physical exertion produces HR elevation and HRV suppression that can be indistinguishable from cognitive arousal without activity-classification context. Ambient temperature variation — proximity to an

extruder barrel at 350–400°F versus ambient shop floor — directly affects cardiovascular load independently of cognitive state. Caffeine, nicotine, and hydration status modulate baseline HR and HRV, creating within-shift and between-shift variability. Social interaction produces arousal signals unrelated to operational decision events.

The mitigation strategy has four components: activity classification from accelerometer data to gate trigger-eligible windows to periods of stationary or low-motion operator state; multi-signal corroboration requirements (convergent deviation across two or more channels — HR, HRV-RMSSD, gaze dwell, or Body Response — within a calibration window before capture fires); continuous personal baseline recalibration that accounts for within-shift drift in cardiovascular baselines; and post-hoc filtering of captured events against known non-decision arousal sources (ambient temperature logs, break schedules, social-interaction timestamps from the audio channel). These confounds and their mitigations are presented as engineering challenges requiring empirical calibration on Line 1, not as solved problems. Whether HR+HRV+motion+gaze without raw cEDA produces an operationally useful trigger — and how it compares to the research-grade EDA baseline available through substitute hardware — is identified in Section 7 as an empirical sub-question for the PPVC Line 1 validation phase.

5.5 The Three-Stage Attention Capture Pipeline

The gaze dwell detection currently deployed in ArcShield is the first layer of a three-stage attention capture pipeline that progressively increases fidelity toward direct measurement of the attentional allocation corresponding to prediction error registration.

Stage 1 — Gaze Dwell Detection (Currently Deployed): Frame differencing on the mobile camera feed detects sustained visual attention on regions of interest, providing the temporal anchor for the Cause phase. Frame differencing in dynamic industrial environments where machinery and other workers are constantly moving will produce elevated false-positive rates; the signal processing approach to distinguish sustained operator attention from ambient visual motion is an active area of calibration. Once Meta Wearables SDK integration is complete, the same frame-differencing algorithm will operate on the glasses' POV camera stream — a better input than any mounted or handheld camera because the frame is genuinely centered on what the operator is looking at.

Stage 2 — Precision Eye Tracking: Tobii Eye Tracker 5, Pupil Labs Neon, or MediaPipe front-camera gaze inference supplements dwell detection with fixation coordinates, saccade velocity, and pupil dilation. Earlier versions of this paper noted that the form-factor of existing precision eye-tracking hardware (desktop-mounted or head-mounted research units) does not integrate naturally into industrial PPE. The Meta glasses platform substantially improves this picture: future generations of AI glasses with integrated eye-tracking optics are publicly on the roadmap, and the Wearables Device Access Toolkit is expected to expose eye-tracking data streams as sensor access matures. Stage 2 integration will track this hardware evolution rather than requiring a custom form factor.

Stage 3 — EEG-Based Neural Ground Truth (Research Aspiration): Consumer EEG devices provide the neural correlate layer for the highest-knowledge-density operators.

This stage is explicitly characterized as a research aspiration rather than a near-term deployment target. Consumer EEG devices in industrial environments face severe motion artifact contamination, electromagnetic interference from motors and VFDs, and sweat-related electrode impedance drift during physical work. The P300 event-related potential detection described here requires controlled stimulus presentation protocols that may be incompatible with free-form industrial operation. Stage 3 validation will require dedicated laboratory-to-field transfer studies.

5.6 Smart Hearing Protection as a Complementary Edge-Compute Node

OSHA mandates implementation of Hearing Conservation Programs whenever workplace noise exposure exceeds 90 dBA as an 8-hour time-weighted average. Smart hearing protection with active noise cancellation, Bluetooth communication, and internal dosimeter sensors via MIRE technology provides a complementary capture channel in environments where OSHA-mandated PPE is already in use. When paired with the AI glasses for visual capture, smart hearing protection adds high-quality acoustic capture of machinery sound, operator speech, and ambient audio without requiring the operator to don any additional device. This configuration is complementary rather than primary: the AI glasses carry the POV camera and the biometric wrist wearable carries the somatic signal, while smart hearing protection carries the acoustic channel in high-noise environments where it is already being worn.

5.7 Connected Worker Market Context

The Connected Worker Platforms market was valued at approximately \$8.6–\$8.9 billion in 2025 (MarketsandMarkets, Mordor Intelligence), with projections reaching \$20–48 billion by 2030–2033 at a CAGR of 18–24%. The Smart PPE segment was valued at \$3.5–\$5.0 billion. Market consolidation — five connected worker companies acquired in 2024–2025 — creates entry opportunity for differentiated architectures.

6. Software Architecture for Bidirectional Reconciliation History Capture

6.1 Sensor Abstraction Layer and Edge AI

The architecture deploys machine learning inference algorithms directly onto local edge devices. Google Health Connect provides the biometric half of the abstraction layer; machine PLC APIs provide the process telemetry half; the Meta Wearables Device Access Toolkit provides the POV camera and audio half. The mobile device functions as the real-time fusion processor maintaining strict time synchronization across all sensor modalities.

6.2 The Knowledge Graph as Bidirectional Reconciliation History

The knowledge graph constructed from accumulated CIAER events is designed to serve as the accumulated bidirectional reconciliation history between a specific agent and a specific

environment: the record of prediction errors registered, inferences performed, actions executed, effects measured, and model updates completed. It is not a database of procedures. Multi-expert divergent chains are preserved as parallel subgraphs. The knowledge graph is partial, in the sense that current instrumentation cannot capture every micro-cycle, but even a partial externalization constitutes a qualitative advance over prior approaches. The full knowledge graph structure is designed architecture, not yet populated at scale; individual CIAER event records are the implemented unit of capture.

6.3 Monte Carlo Augmentation of Real Reconciliation Chains

A small initial corpus of real CIAER events can be expanded through physically constrained Monte Carlo augmentation to provide training volume for preliminary model development. The augmentation operates exclusively on the structured, sensor-derived components of each record — time-series biometric signals, process parameter distributions, and temporal phase relationships — rather than on voice annotation content, which is never fabricated. All augmented data is explicitly labeled and distinguished from real observations. A held-out partition of purely real, un-augmented events is maintained exclusively for validation and testing.

6.4 Causal Structure, Behavioral Cloning, and Toward Thought Cloning

Section 2.16 argued that pure behavioral cloning of expert behavior is pathologically susceptible to causal confusion, and that the CIAER five-phase decomposition supplies — by construction — the causal DAG of each decision event. This section describes the downstream implication: the corpus that CIAER builds is designed to be a native substrate for causally-grounded training objectives that conventional demonstration datasets cannot support.

The cleanest articulation of the needed training objective is Hu and Clune’s Thought Cloning framework (arXiv:2306.00323, NeurIPS 2023), which trains AI systems not merely on behavioral actions but on the synchronized thought processes underlying those actions. Thought Cloning in its original form was validated only with synthetic thought data in the BabyAI grid environment; its authors explicitly identified the need for real-world corpora of humans “thinking out loud while acting” as the bottleneck to demonstrating its full potential. ArcShield’s corpus is designed precisely to supply that bottleneck resource: every captured event pairs an environmental Cause, a voice-annotated-or-biometric-signed Intuition, an observed Action, a measured Effect, and an evaluated Result — which is, structurally, an industrial-domain instance of the synchronized action-plus-reasoning trace that Thought Cloning requires as training input.

The five-phase decomposition also addresses the specific causal-confusion pathologies that de Haan et al. (2019) and Ortega et al. (2021) diagnose. Because each captured event separates Cause (the specific prediction error the expert responded to) from Action (what the expert did in response), a downstream model trained on this corpus has structured access to the variable the expert was conditioning on rather than merely to the set of variables present in the sensor feed at action time. This converts the causal inference problem from one of latent-structure discovery — where de Haan et al. show that

behavioral cloning systematically fails — to one of supervised conditioning on a known causal variable. Ortega et al.’s meta-learning approach, Ruan et al.’s instrumental-variables approach, and Shao et al.’s double-machine-learning approach all become available as training-time options against a corpus structured this way; none of them would be usable against a raw behavioral-log corpus.

The structural correspondence extends to the Effect and Result phases. Behavioral cloning conventionally trains against state-action pairs, with no native mechanism for weighting training examples by the outcome they produced. The CIAER corpus natively pairs each action with its measured effect and the expert’s evaluation of that effect against prediction — which is the supervisory signal that causal imitation learning and outcome-weighted behavioral cloning require. A confirming Result supplies a positive-example weighting; a disconfirming Result supplies either a negative-example weighting or, more usefully, an explicit model-update record that downstream training can consume as a curriculum signal. Multi-expert divergent chains (Section 3.4) provide an additional layer of supervisory signal: where two experts observing the same Cause produce different Intuition-to-Action paths, the Result on each becomes a direct empirical discriminator between the two expert models.

Two honest caveats apply. First, the capture architecture does not eliminate causal confusion in the trained downstream model; it supplies the structural prerequisites that make causal training objectives feasible. Whether a specific training run recovers the intended causal structure depends on the architecture of that training run, the size and diversity of the corpus, and the quality of the Cause-annotation. Second, the pathology that Pezzulo et al. (2024) identify — that passive capture systems inherently access only behavioral correlates of embodied cognition, not the embodied cognition itself — applies to the ArcShield architecture as stated. We accept this critique and address it directly in Section 9: the biometric layer, voice annotation, and gaze capture access physiological, verbal, and attentional *correlates* of expert cognition. They do not access expert cognition directly because that access does not exist. The question the validation program (Section 7) is designed to answer is whether the correlates, captured at the resolution CIAER prescribes, are sufficient to train downstream models that replicate the expert’s decision function at useful fidelity. This is an empirical question that the current proof of concept does not answer and the expansion program is designed to.

6.5 Human-in-the-Loop Validation

The architecture permanently embeds Human-in-the-Loop validation. Every accepted recommendation reinforces the corresponding knowledge graph pathway’s confidence weighting; every rejected or modified recommendation logs a new CIAER event, updating the graph to reflect variables the algorithm missed.

One caveat from the human-AI teaming literature requires acknowledgment. Schmutz et al. (2024) and related work have documented that introducing AI teammates into human work teams can reduce team coordination, communication, and trust, particularly when the AI’s recommendations are opaque or perceived as intrusive. This is a known risk for any human-in-the-loop deployment and ArcShield is not exempt. The architectural response is

that recommendations are explicitly framed as suggestions drawn from the operator’s own captured reconciliation history and the captured history of designated peer experts — not as authoritative prescriptions from an opaque model. Operators see which CIAER events in the graph support a recommendation and retain full authority to override. The validation deployment (Section 7) will treat trust and coordination effects as measurable outcomes rather than secondary concerns.

7. Multi-Domain Validation: The Single-Facility Advantage

The CIAER hypothesis predicts that the schema will manifest with structural invariance across every domain where embodied, experienced agents act purposefully in dynamic environments. Validating this prediction requires operational environments that differ in sensory modalities, decision timescales, and physical feedback characteristics. A single manufacturing facility housing multiple operationally distinct departments provides a methodologically superior validation design: genuine cognitive domain diversity under controlled organizational conditions, eliminating confounding variables that would complicate attribution in a multi-facility study.

The validation design draws methodological inspiration from Verkerk et al. (2026), who demonstrated that proposed linguistic universals must be tested while controlling for non-independence arising from shared ancestry and geographic proximity. In our context, operators trained at the same facility share a “knowledge genealogy” analogous to linguistic descent, and neighboring departments share process culture analogous to geographic diffusion. The single-facility design controls for organizational culture, shift structure, and management practices while providing genuine cross-domain variation in cognitive demands — an approach structurally analogous to the spatiophylogenetic controls employed in comparative linguistics.

7.1 PVC Extrusion — Proof of Concept, Currently Deployed

PPVC Line 1 serves as the foundational proof of concept with dominant Knowledge-based and Rule-based SRK profile and 15–60 second decision timescales. Initial biometric baseline calibration and CIAER JSON schema definition are established here before expansion. The biometric integration layer is in calibration phase establishing personal HR and HRV-RMSSD baselines and training the activity-classification gating model.

A specific empirical sub-question of this deployment concerns the HRV-centric trigger architecture defined in Section 5.2. The theoretical framework (Damasio, §2.14) most directly predicts that EDA would carry the richest somatic-marker signal, but raw cEDA is not exposed by the consumer-wearable platform. The Line 1 deployment therefore tests whether HR + HRV-RMSSD + accelerometer-gated activity classification + gaze dwell — the channels that *are* exposed — produce an operationally useful trigger for CIAER event capture in a live industrial environment. A complementary validation sub-study is planned in which a subset of operators wear a research-grade EDA device (such as the Empatica EmbracePlus) alongside the Pixel Watch 4 during matched shifts. Comparing the trigger events produced by the two configurations provides a direct empirical test of the value

added by raw EDA over the HRV-based trigger. If raw EDA provides substantial additional discrimination, that finding justifies a Gen 3 hardware procurement decision favoring research-grade wearables for operationally critical deployments; if the HRV-based trigger is sufficient, the finding retires a speculative dependency and simplifies the deployment architecture. This sub-question is identified as high-priority and is expected to be resolved within the Line 1 event corpus.

7.2 Injection Mold — Highest-Value Expansion Target

The most direct parallel to extrusion in polymer processing expertise, with complex tool-material interaction generating the highest proportion of Knowledge-based Variation 3 events. Combined with the extrusion corpus, the injection mold dataset provides within-category schema consistency validation for polymer processing.

7.3 Sheet Metal Fabrication — Thermomechanical Forming Domain

Introduces a distinct process vocabulary centered on material behavior interpretation. Well-suited for demonstrating the biometric somatic marker layer's value at the Rule-based level.

7.4 Filter Assembly — Skill-Based Domain and Divergence Test Venue

The dominant Skill-based SRK profile shifts expert knowledge into the kinesthetic and tactile domain. Multiple operators performing identical tasks simultaneously with measurable downstream quality outcomes constitutes the designed empirical venue for multi-expert divergence validation.

7.5 Packaging — High-Volume Reconciliation Corpus

The highest-volume decision event department, providing the statistical base for Monte Carlo augmentation calibration and biometric trigger threshold optimization.

7.6 Shipping and Receiving — Non-Physical Information-Processing Domain

The only information-processing domain in the manufacturing portfolio, where prediction errors are documentary and relational rather than physical. This domain challenges the biometric capture layer most severely: there is no strong prior reason to expect that HR and HRV deviations during document-based anomaly detection will have the same magnitude or reliability as those during physical process anomaly detection. If the CIAER schema fails to describe expert action in this domain, the minimum sufficient schema claim would require qualification. If it succeeds, the finding extends the hypothesis beyond the physical world into the information domain. The somatic marker hypothesis was developed for embodied decision-making under physical uncertainty; its extension to clerical information processing is a further theoretical stretch that the validation design is specifically constructed to test. The fast-food shift-management instance described in Section 4.6 represents a structurally analogous abstract-environment domain outside the manufacturing context; the convergence or divergence of CIAER patterns between

Shipping/Receiving and an analogous service-operations deployment would provide cross-sector evidence on whether the schema generalizes beyond physical embodiment.

8. Competitive Landscape Analysis

The architecture described above raises a natural question: do existing connected worker platforms already capture decision logic at this level of structural detail? Based on publicly available product documentation and announced capabilities as of early 2026, a review of major connected worker platforms — Squint AI, Augmentir, Tulip, DeepHow, InSkill, Dozuki, Parsable, PTC Vuforia, Cognite, Oppr.ai, and Interloom — finds that no reviewed platform employs a structured causal decision schema comparable to the CIAER architecture. Every platform examined documents the Action phase — what workers do — without systematically capturing the Cause that initiated the action, the Intuition inference that selected it, the Effect it produced, or the Result that updated the agent’s model.

ArcShield’s Three Defensible Differentiators (based on public documentation as of early 2026): (1) the CIAER schema is the only published framework structuring captured knowledge as a closed bidirectional Cause-Intuition-Action-Effect-Result loop with SRK-level tagging and multi-expert divergence preservation; (2) no competitor integrates knowledge capture into commodity AI glasses as a POV platform or incorporates a physiological somatic marker layer for passive Intuition phase detection; (3) no competitor offers a zero-IT-footprint, zero-marginal-cost-per-event, single-file deployment architecture.

A further structural difference concerns the technical frame of the capture problem. Connected worker platforms in the reviewed set frame their task as *procedure documentation and guidance*: translating existing SOPs into interactive digital formats and surfacing contextual help to workers executing defined procedures. This frame takes the propositional SOP as the ground truth and the worker as an executor of it. The CIAER frame is structurally inverse: the expert’s non-propositional generative model is the ground truth, the SOP is a degraded projection of that model, and the capture objective is to externalize the model directly rather than to improve the SOP projection. The causal-confusion argument of Section 2.16 is the technical articulation of why this inversion matters: an architecture built on the procedure-documentation frame inherits the pathology of behavioral cloning by construction, because a procedural log is exactly the input that behavioral cloning consumes and on which it confuses correlates with causes. An architecture built on the bidirectional-reconciliation frame can, in principle, avoid this pathology because it captures the causal variables the expert conditioned on, not merely the behavioral trace those variables produced. Whether the theoretical advantage translates into a measurable downstream-model quality advantage is the empirical question that the validation program (Section 7) is designed to resolve.

9. Strategic Implementation and Proof of Concept

The multi-domain validation roadmap above is designed to test the CIAER hypothesis at scale. The current proof of concept, ArcShield, validates the foundational pipeline on the first of these six departments.

ArcShield is implemented as a single-file progressive web application utilizing computer vision for instrument reading, Web Speech API for operator voice capture, gaze dwell detection via frame differencing, and biometric event triggering via Google Health Connect integration on a Pixel 9 Pro device, capturing the complete five-phase decision schema at zero marginal cost per event on a live PVC extrusion line at a manufacturing facility in Helena-West Helena, Arkansas. The zero-IT-footprint, single-file architecture serves as empirical proof that the complete CIAER pipeline can be deployed and validated on a live production floor without enterprise IT involvement, capital expenditure, or worker adoption friction. The Pixel 9 Pro deployment is the interim capture platform; migration to the Meta AI glasses platform described in Section 5.1 is the production target as the Wearables Device Access Toolkit matures toward general availability.

9.1 Current Deployment Status (April 2026)

As of April 2026, the ArcShield proof of concept has achieved the following milestones. The CIAER five-phase JSON schema is defined and functional, generating structured event records during active production on PPVC Line 1. The gaze dwell detection module is operational, using frame differencing on the smartphone camera to identify sustained visual attention. Voice annotation capture via the Web Speech API has been validated in test sessions, producing natural-language operator annotations structured into the five-phase schema. Computer vision gauge reading via Claude and Gemini Vision APIs has been tested and validated for analog instrument digitization. The biometric integration layer is in baseline calibration phase, establishing personal HR and HRV-RMSSD baselines from the Google Pixel Watch 4 via the Health Connect API, and training the accelerometer-based activity-classification gating model. The system is early-stage; the current corpus is small and the deployment is best characterized as foundational validation of the capture pipeline rather than comprehensive empirical proof of the CIAER hypothesis.

10. Limitations and Future Directions

This section describes the principal limitations of the work as it currently stands, organized to correspond to the six categories surfaced by the April 2026 literature review and stress-testing process.

10.1 The Central Hypothesis Requires Direct Empirical Validation

The proposition that CIAER describes a candidate minimum sufficient behavioral schema for adaptive purposeful behavior in embodied agents is grounded in convergent theoretical testimony and consistent with neurobiological findings, but it has not been validated through direct experimental confirmation in the industrial context. The six-department

deployment roadmap constitutes the designed experimental infrastructure for this validation. The hypothesis is advanced as the most parsimonious explanation for the eleven-framework convergence, not as a proven fact. The minimality of the decomposition is a testable prediction, not a demonstrated property.

10.2 The Empirical Status of Predictive Coding and the FEP

Section 2.13 characterizes the CIAER-FEP mapping as functional isomorphism, not formal derivation. A related caveat is that the broader predictive-coding program within which active inference sits has accumulated converging but still modest empirical support — strong in specific regions of visual cortex, more tentative elsewhere. This does not undermine the CIAER framework because the framework is advanced at the behavioral level of description, not at the level of claims about the specific neural implementation of prediction error processing. We flag the empirical status of predictive coding because a reader interested in the neural-substrate claims that sometimes travel alongside active inference discourse should not assume those claims inherit the status of CIAER as a behavioral schema.

10.3 The Passive-Capture / Embodied-Cognition Critique

Pezzulo et al. (2024) and related work in the active-inference literature argue that passive AI systems — systems that observe behavior without engaging in it — cannot access the embodied understanding that active agents possess. A sharper form of this critique applied to ArcShield is that any digital capture system necessarily records behavioral correlates of embodied cognition rather than the embodied cognition itself, and that those correlates are a fundamentally inadequate substrate for reconstructing the cognition. This critique is also related to the causal confusion literature’s recursive worry: if capturing causal structure is central, passive sensor capture only ever accesses behavioral correlates of causal reasoning, not the reasoning itself.

We accept the critique as stated and answer it with the following position. The CIAER capture architecture does not claim to access expert cognition directly; it accesses physiological, verbal, and attentional correlates of expert cognition, structured into a causal decomposition that makes downstream training objectives feasible. Whether the structured correlates, at the sampling resolution and decomposition granularity the architecture provides, are sufficient to train downstream models that replicate expert decision functions at useful fidelity is an empirical question that the validation program is designed to answer. The critique is therefore not a reason to abandon the architecture; it is a reason to design the validation program so that its success criterion is measured downstream-model performance rather than intuitive completeness of the capture. The current proof of concept does not answer the question; the multi-department deployment with held-out validation events is designed to.

10.4 The Dreyfus-Peña Critique of Pure Intuition

Peña (2010) argued that even expert practitioners alternate between intuitive and analytical modes rather than operating purely from intuition. We accept this refinement to

the Dreyfus model and integrate it into the CIAER framework rather than against it, as described in Section 2.7: Klein’s three RPD variants are the topological correlates of Peña’s mode-alternation within the Intuition phase, and the SRK classification of each captured event explicitly tags the cognitive mode in operation. Peña’s critique does not undermine the framework; it constrains the framework to support all three modes, which it does. The limitation that remains is practical: the current proof of concept does not yet have sufficient event volume to demonstrate coverage across all three modes, and the expansion program must explicitly validate that high-SRK-knowledge events are captured with fidelity rather than only high-SRK-skill events.

10.5 The Somatic Marker Extension Requires Industrial Validation

The application of the somatic marker hypothesis to industrial expert judgment is a theoretical extension requiring its own empirical confirmation. Overskeid (2021) and related recent critics have argued that the SMH literature has accumulated stronger interpretations than Damasio’s own evidence supports. Our use is deliberately restricted to the narrower claim Damasio himself defends: that somatic signals correlate with and bias decision-making under uncertainty in ways detectable through physiological measurement. We do not claim somatic markers uniquely determine decisions or that the biometric layer accesses the content of expert cognition directly. Known confound sources in industrial environments — physical exertion, ambient temperature variation, caffeine and hydration status, social interaction — are acknowledged in Section 5.4, with planned mitigation strategies.

A distinct and important limitation concerns the gap between theoretical prediction and platform reality. Damasio’s framework most directly predicts that EDA would carry the richest somatic-marker signal, but raw cEDA is not exposed by the consumer wearables on which the current deployment tier runs. The operational trigger therefore rests on HR, HRV-RMSSD, accelerometer-gated activity classification, and gaze dwell. Whether this combination is a sufficient operational proxy for the EDA-centric signal Damasio’s framework nominates — or whether an accuracy penalty is paid for the substitution — is an open empirical question specifically assigned to the Line 1 validation phase (Section 7.1). Pre-submission, this gap is a known limitation of the paper: the theoretical grounding points at EDA; the deployed system triggers on HRV. Resolution awaits empirical data.

10.6 The Human-AI Teaming Trust Problem

Schmutz et al. (2024) and related work have shown that introducing AI teammates into human work teams can reduce team coordination, communication, and trust. ArcShield is not exempt from this risk. The architectural response is framed in Section 6.5: recommendations are explicitly drawn from the operator’s own captured history and from peer experts designated as trusted sources, rather than from an opaque model; operators see the CIAER events supporting any recommendation and retain full override authority. Whether this framing actually prevents trust erosion in practice is an empirical question the multi-department deployment will measure. Trust and coordination effects are treated as primary outcomes alongside capture fidelity, not as secondary concerns.

10.7 Voice Capture Accesses Only Conscious Processing — Partially Addressed

Per Ericsson and Simon’s protocol analysis research, verbalization only accesses processing in short-term memory. The biometric somatic marker layer provides indirect evidence of the bidirectional loop for Skill-level events even when no verbal report is forthcoming. The two-phase capture architecture — silent background logging during the shift with asynchronous conversational elicitation afterward — is the current response to the real-time verbalization impossibility, but its efficacy in capturing Skill-level content that the operator could not articulate in the moment remains an empirical question.

10.8 Single-Facility Proof of Concept

The six-department portfolio provides genuine cross-domain validation under controlled organizational conditions. Future work should extend to multi-facility comparison, controlling for shared training lineage and organizational culture using methods analogous to the spatiophylogenetic controls demonstrated by Verkerk et al. (2026) in comparative linguistics. The fast-food shift-management instance described in Section 4.6 points to a natural cross-sector expansion target: a service-operations deployment that tests CIAER structural invariance outside manufacturing entirely, against an abstract ratio-dashboard environment rather than a physical process.

10.9 EEG and Operational Pupillometry are Research Aspirations

Stage 3 of the attention capture pipeline — consumer EEG in industrial environments — faces severe practical challenges including motion artifacts, electromagnetic interference, and sweat-related electrode drift. This stage is characterized as a long-term research target, not a near-term deployment plan. Direct pupillometric tracking during operations would constitute the most specific empirical test of the anticipatory-inference hypothesis advanced in §2.17, but it requires inward-facing eye-tracking capability on the glasses platform that current-generation hardware does not yet expose to third-party developers. Pupillometric integration therefore shares the research-aspiration status of the EEG stage and is explicitly out of scope for the current proof-of-concept deployment.

10.10 Privacy and Regulatory Compliance

The EU AI Act classifies workplace AI systems used for individual performance monitoring as high-risk with extensive compliance requirements beyond data localization, including conformity assessments, technical documentation, human oversight provisions, and transparency obligations to workers. The on-device processing architecture addresses data protection requirements, but full high-risk system compliance requires additional measures. The characterization of the system as safety documentation infrastructure rather than performance monitoring has not been tested against regulatory interpretation; this interpretive ambiguity is acknowledged.

10.11 Competitive Analysis Based on Public Documentation

Internal product architectures of competing connected-worker platforms may contain capabilities not publicly documented. The competitive-differentiation claims in Section 8 are grounded in publicly available product documentation and announced capabilities as of early 2026 and should be read with that scope in mind.

10.12 Graph Weight Reflexivity and the Need for Withhold-Sampling

A subtler failure mode arises once the system’s accumulated knowledge graph begins influencing operator behavior at deployment scale. The CIAER graph assigns weights to reconciliation pathways based on confirmed-Result outcomes — pathways whose predictions were empirically validated accumulate weight; disconfirmed pathways lose weight. This updating rule is appropriate during the initial corpus-construction phase, where operators act from their own generative models and the graph merely records outcomes. Once the Personal AI Twin begins offering advisory guidance derived from the graph, however, a reflexivity problem emerges: if operators follow high-weight guidance, the resulting actions will tend to produce confirming Results, which will further reinforce the weights, which will produce more confirmations — even in regions of the operational envelope where the underlying causal model is wrong or has drifted. The graph becomes a self-confirming artifact rather than an independently validated model of the domain. This is structurally identical to the feedback-loop pathology documented in algorithmic prediction systems from recommender systems to predictive policing, and it is a known failure mode of reinforcement learning policies deployed in environments they are also shaping.

The intended mitigation is a withhold-sampling policy applied at the guidance-delivery layer. At an empirically calibrated rate, the Twin withholds high-weight guidance on a randomly selected subset of matched events, allowing the operator to act from their own model unassisted. The resulting Cause-to-Result chains for those events enter the graph as *counterfactual samples* — observations of what would have happened without the Twin’s influence — and are used to detect divergence between the graph’s current prediction and operator-driven outcomes. Persistent divergence on withheld samples is the signal that the graph’s current weights no longer reflect operational reality and that re-weighting or re-training is required. The sampling rate, the selection policy, and the threshold for triggering re-weighting are all empirical parameters requiring calibration during deployment. This policy is not yet implemented in the current proof-of-concept architecture; it is documented here as a known architectural requirement for any scaled deployment and is a prerequisite for responsible Personal AI Twin guidance beyond the initial corpus-construction phase. The marketplace-layer implications of this requirement are elaborated in the companion product architecture documentation.

AI Disclosure

This work was developed with the assistance of large language model tools, specifically Anthropic Claude (models accessed between January and April 2026), which was used for research synthesis, literature organization, competitive landscape analysis, structural

editing, and iterative drafting of sections. All theoretical frameworks, domain expertise, operational observations, and core conceptual contributions originate from the author. These contributions include the five-phase CIAER decision event schema; the hypothesis that CIAER describes a candidate minimum sufficient behavioral schema for adaptive purposeful behavior in embodied agents; the bidirectional reconciliation loop formalization; the synthesis of convergent theoretical frameworks into a unified argument; the integration of the causal confusion literature as a technical validation of the framework’s externalization approach; the extension of Damasio’s somatic marker hypothesis to industrial expert judgment; the three-stage attention capture pipeline; the AI-glasses-as-data-platform hardware strategy; the multi-domain single-facility validation design; the Monte Carlo augmentation methodology; the on-device privacy architecture; and the ArcShield proof-of-concept design. AI assistance was employed as a research and writing aid; all factual claims, citations, and technical assertions were reviewed and verified by the author. The author takes full responsibility for the accuracy and integrity of all content. This disclosure is provided in accordance with arXiv’s AI-assisted content policy.

Suggested arXiv primary classification: cs.HC. Cross-list: cs.AI.

Methodological Note: Adversarial Stress-Testing of the Framework

The theoretical commitments in Section 2.1 — the topological justification for five phases, the multi-timescale tolerance of the phases, and the two-level scope criterion combining functional self-maintenance with evolutionary grounding — were not arrived at in a single analytical pass. They are the product of an iterative adversarial stress-testing process in which candidate counterexamples were systematically generated against earlier versions of the framework, and each successful challenge prompted a refinement rather than a rescue. We document the lineage of this reasoning here for two reasons: first, to establish that the framework’s foundational claims have been probed from multiple angles rather than merely asserted; second, to give future critics a record of which objections have already been considered, so that subsequent debate can advance the theoretical conversation rather than rediscover settled ground.

The initial version of the paper asserted CIAER as “the minimum sufficient schema for adaptive purposeful behavior in embodied agents” without formal minimality proof, without topological justification for the phase count, without tolerance for heterogeneous phase timescales, and without an explicit scope criterion for embodied agency. A series of candidate counterexamples progressively exposed the weaknesses of this formulation and drove its refinement.

The Venus flytrap. At first reading, a four-phase agent (Cause, Action, Effect, Result) apparently lacking Intuition. Resolved by recognizing that the flytrap’s calcium ion integration across multiple trigger hairs, requiring two deflections within roughly 20–30 seconds, is a genuine discriminative computation distinguishing live prey from raindrops and debris. The Intuition phase is present, implemented in non-neural substrate, operating on a sub-second timescale. This case forced the clarification that Intuition is substrate-neutral: any computation transforming a registered prediction error into an

action selection qualifies, regardless of whether it is implemented in cortical tissue, calcium ion dynamics, or silicon.

Bacterial chemotaxis (*E. coli*). The strongest candidate for a genuinely four-phase agent. Cause, Action, Effect, and Result (receptor methylation adaptation) are clearly present, but Intuition — as phosphorylation of CheY proportional to receptor activity — is close to a direct proportional transduction with no discriminative work. This case could not be cleanly rescued by appeal to rich internal computation, but it illustrated that phase sophistication varies across a continuum from near-trivial to elaborate. The framework accommodates chemotaxis by granting that Intuition may be computationally minimal in some agents, while noting that topological presence of the phase is distinct from computational sophistication within it.

The bimetallic-strip thermostat. The *reductio ad absurdum* test. A negative feedback control system with Cause (temperature detection), Action (circuit closure), Effect (temperature change), and arguably Result (feedback to the sensor), but lacking any experience-dependent update mechanism at any timescale. Acknowledged as outside the framework's scope. This case drove the addition of the experience-dependent update requirement to the scope criterion: without some mechanism of model update, however slow, a system is a fragment of a CIAER loop rather than a complete one. Pure fixed-function transducers are explicitly excluded.

The patellar reflex. Dewey's founding example, and therefore an embarrassing case if it did not fit the framework. Initially appeared to lack both Intuition and Result: the monosynaptic arc is hardwired and performs no learning within a single stimulation event. Resolution required two moves. First, the alpha motor neuron's leaky integration with threshold crossing was acknowledged as a minimal-but-genuine Intuition phase, implementing temporal integration of graded input into a binary action decision. Second, the reflex's Result phase was located in slower mechanisms: gamma motor neuron gain modulation adjusts spindle sensitivity across seconds to minutes, descending inputs from higher centers modulate the reflex across behavioral episodes, and long-term plasticity alters reflex thresholds across years. This case drove the addition of multi-timescale tolerance: phases need not operate on a single clock, and an inward arc may be implemented on a much slower timescale than the corresponding outward arc through entirely separate anatomical or computational channels.

The rolling rock. The unfalsifiability probe. Any physical process with feedback can be force-fit into five phases through sufficiently loose readings of "inference" and "model update." This case demonstrated that the framework's discriminative content cannot come from the phases themselves, since the phases can be parsed onto almost any physical system. The criterion that does the real work of distinguishing agents from non-agents is the agent-environment bidirectional reconciliation requirement combined with experience-dependent update. The rock is not reconciling anything — it has no self-maintenance orientation, no mechanism for preserving its own organization — and is therefore outside the framework's scope regardless of how the phases might be mapped onto its physics.

The chess grandmaster. A case that appeared to have more than five phases: generation of candidate moves, evaluation through mental simulation, selection among candidates, and only then Action. Resolution required the topological argument: generation, evaluation, and selection all occur on the same side of the agent-environment boundary, between receiving the position and committing to a move. They are rich internal structure within the Intuition phase, not separate phases, because none crosses a new boundary. This case drove the explicit articulation that phase count is determined by topology and boundary-crossings rather than by enumeration of sub-operations, and that Klein's three RPD variants are topological variants of the Intuition phase rather than distinct phase structures.

The sterile expert. A probe of the initial procreation-based scope criterion. If embodied agency required biological reproductive capacity, the framework would exclude post-reproductive master craftsmen, childless experts, and vasectomized welders — exactly the population the industrial capture use case targets. This case drove the distinction between proximate self-maintenance (which the individual possesses regardless of reproductive status) and evolutionary grounding (which the individual inherits from the lineage that built them). The sterile welder is unambiguously an embodied agent because his agency was established by the reproductive history that produced his current organization, not by his individual reproductive status.

The AI system trained on expert data. A probe of whether biological grounding excludes artificial agents from the framework. Under strict Millikan teleosemantics, an AI system has no original intentionality because it has no reproductive history of its own. This would render the Thought Cloning direction in Section 6.4 theoretically incoherent. Resolution required broadening “procreation” to its etymological sense of forward-directed production under selection, and distinguishing original intentionality (biological, evolutionary) from derivative intentionality (artificial, design-mediated). AI systems participate in CIAER derivatively through their training history, which implements variation and selection over weight updates under objectives supplied by biological designers whose own agency is original. The framework applies to both, but with different grounding stories. This distinction is theoretically consequential and commercially irrelevant: the practical goal of transferring expertise from operator to AI is unaffected by the philosophical observation that the two agencies have different natural histories.

The fast-food shift manager. A probe of whether the framework's structural invariance extends beyond physically embodied processes into abstract ratio-based operational environments. A case where the “environment” is a dashboard of labor percentage, sales per man-hour, order times, and staffing-slot projections rather than any physical process, and the “action” is a discrete staffing intervention rather than a mechanical adjustment. This case drove the explicit articulation that the Cause-Intuition-Action-Effect-Result topology is preserved across the physical-to-abstract domain shift: the five phases retain their identity, their boundary-crossings, and their capture requirements even when the environment being reconciled against is an informational construct. Section 4.6 develops this case in detail.

Each of these cases produced a refinement rather than a collapse, but the refinements were substantive rather than cosmetic. The topological justification for five phases, the

multi-timescale tolerance, the experience-dependent update scope criterion, the two-level functional-plus-grounding criterion for embodied agency, and the domain-agnostic structural invariance across physical and abstract environments were all absent from the initial formulation of the framework and were added in response to specific challenges. The framework that survives this process is narrower and more precisely specified than the framework that entered it, and it is defensible on terrain where the earlier version was not. We regard this stress-test lineage as part of the contribution of the paper: it establishes that the framework's foundational commitments have been probed from multiple angles rather than merely asserted, and it provides a map of the terrain on which future critics can advance the theoretical conversation rather than rediscover objections already considered and addressed.

We note that the adversarial stress-testing process was conducted iteratively with large language model assistance, specifically Anthropic Claude, serving as a rigorous interlocutor generating candidate counterexamples and pressing hard on weaknesses in each successive formulation. The theoretical commitments and their justifications originate from the author; the counterexample generation and the pressure-testing of each proposed resolution were performed collaboratively. We document this explicitly because we believe the resulting framework is stronger than it would have been without the stress-test, and because we want to encourage this kind of adversarial theoretical work as a standard practice in preprint development rather than treating it as an incidental feature of the drafting process.

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